

Riccardo Buscicchio | Curriculum Vitæ

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Academic positions

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|---|---------------------------------------|
| Gran Sasso Science Institute
<i>Tenure-track assistant professor</i> | L'Aquila, Italy
2026 - 2032 |
| Università degli Studi di Milano-Bicocca
<i>Postdoc (Assegnista di ricerca), Department of Physics "G.Occhialini"</i> | Milan, Italy
2021 - 2024 |
| ◦ Main activity: development of LISA data analysis ground-segment for the Italian Space Agency (Phase A). | |
| Università degli Studi di Milano-Bicocca
<i>Postdoc (Assegnista di ricerca), Department of Physics "G.Occhialini"</i> | Milan, Italy
2024 - current |
| ◦ Main activity: development of LISA data analysis ground-segment for the Italian Space Agency (Phase B). | |

Education

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|---|--|
| University of Birmingham
<i>Ph.D., School of Physics & Astronomy</i> | Birmingham, UK
2017-13/07/2022 |
| ◦ Supervisor: A. Vecchio. Thesis resulted in 6 short-author publications. | |
| ◦ Thesis Title: Topics in Bayesian population inference for Gravitational Wave Astronomy | |
| Università degli Studi di Pisa
<i>Master's degree in Theoretical physics</i> | Pisa, Italy
2013-2016 |
| ◦ Supervisor: G. Cella. Thesis resulted in one short-author publications. | |
| ◦ Thesis title: An improved detector for non-gaussian stochastic background of gravitational waves. | |
| Columbia University
<i>INFN-NSF Summer Internship</i> | New York NY, USA
Jun-Sept 2013 |
| ◦ Supervisor: S. Marka, I. Bartos. | |

Metrics

Publications:

- 35 short-author papers published in major peer-reviewed journals
(out of which 8 first-authored papers and 9 lead by supervised student).
- 14 collaboration papers with substantial contribution, published in major peer-reviewed journals
- 60 collaboration papers in total, published in major peer-reviewed journals
- 8 papers in submission stage,
- 2 other publications (thesis, white papers, reviews)

Total number of citations: >18800. h-index: 27 (from ADS and INSPIRE record).

Web links to list services: [ADS](https://arxiv.org); [INSPIRE](https://inspirehep.net); [arXiv](https://arxiv.org); [orcid](https://orcid.org).

Full list of publications available at www.riccardobuscicchio.com/publications.

Full list of presentations available at www.riccardobuscicchio.com/talks.

Codes & Datasets

Title	Code	Dataset	Zenodo DOI	Public
○ Bahamas Galactic Foreground	✓	✓	10.5281/zenodo.17527235	✓
○ Bahamas	✓		10.5281/zenodo.16087705	✓
○ Hypertriangulation Map	✓		10.5281/zenodo.13897708	✓
○ Bayesian PowerLaw Sensitivity	✓	✓	10.5281/zenodo.14384633	✓
○ Milky Way Satellites		✓	10.5281/zenodo.3668904	✓
○ LISA stellar BBH catalogues and samples	✓	✓	10.5281/zenodo.14426778	✓
○ LISA MBHB catalogues and samples		✓	10.5281/zenodo.13787674	✓
○ Chirp Gravitational Wave Alerts	✓		10.5281/zenodo.3525063	✓

Grants, Prizes, & Awards

Career prizes:

- Braccini PhD Thesis Prize, Gravitational Wave International Committee honorable mention. 2021
- Michael Penston PhD Thesis Prize, Royal Astronomical Society runner-up prize. 2021

Grants:

- Royal Society of Edinburgh “Gravity under the lens” (col, PI Dr. Berry, Univ. of Glasgow, UK) 2025-2027
- EuroHPC PRACE “LABELFIT” proposal, 224k CPUh on Leonardo DCGP 2025
- EuroHPC PRACE “LISA-FIT” proposal, 100k CPUh on Leonardo BOOSTER 2023
- Google Cloud for Researchers, 4kEUR Google Cloud Research Credits 2023
- CINECA ISCRA Type-C project “LISA-MW” proposal, 10k CPUh at the Italian National HPC center. 2022
- Travel Grant, Horizon 2020 AHEAD 2020 (High Energy Astrophysics) 2021
- Travel Grant, American Physical Society, DGRAV Student Travel Grant 2020
- Travel Grant, Institute of Physics Student Travel fund 2019
- Travel Grant, Royal Astronomical Society, UK. 2018

Invited research visits:

- Kavli visitor program research visit — DAMTP Univ. Cambridge, Cambridge, UK 07/10/2024-18/10/2024
- Research visit — Max Planck Institute for Astrophysics, Garching, DE 23/11/2023-30/11/2023
- Research visit — Albert Einstein Institute, Potsdam, DE 16/04/2023-23/04/2023
- Research visit — Univ. of Stavanger, Stavanger, NO 08/02/2023-29/03/2023
- AHEAD 2020 program research visit — APC, Univ. Paris Diderot, Paris, FR 01/12/2023-07/01/2023
- Research visit — APC, Univ. Paris Diderot, Paris, FR 26/10/2022-30/11/2022
- Research visit — Univ. of Stavanger, Stavanger, NO 26/09/2022-19/10/2022
- Research visit — Univ. of Birmingham, Birmingham, UK 05/07/2022-10/07/2022
- Research visit — Columbia University, New York, US 06/04/2022-13/04/2022

Student supervision

According to current national regulations, as a research fellow I cannot be officially appointed as supervisor of students at any level. However, upon agreement with the relevant permanent staff, I have supervised the work of students in the percentages shown below.

Ph.D. student co-supervisor:

- B. Saiguhan, University of Milano-Bicocca, PhD, 100% 2025-2028
- L. Viganò, University of Milano-Bicocca, PhD, 80% 2025-2028
- F. Nobili, University of Insubria, PhD, 100% 2024-2027
- A. Spadaro, University of Milano-Bicocca, PhD, 100% 2022-2025
- F. Pozzoli, University of Insubria, PhD, 100% 2022-2025

MSc student co-supervision:

- R. Bosoni De Martini, University of Milano-Bicocca, Master's thesis, 100% 2025-2026
- L. Viganò, University of Milano-Bicocca, Master's thesis, 100% 2024-2025
(now PhD student at Univ. of Milano-Bicocca)
- M. Bellotti, University of Milano-Bicocca, Master's thesis, 100% 2024-2025

(now PhD student at Observatoire Cote d'Azur)	
○ D. Chirico, University of Milano-Bicocca, Master's thesis, 100%	2023-2024
○ S. Corbo, Politecnico di Milano, Master's thesis, 100%	2023-2024
○ R. Rosso, University of Pisa, Master's thesis, 80%	2023-2024
(now PhD student at Univ. of Pisa)	
○ G. Astorino, University of Pisa, Master's thesis, 80%	2023-2024
(now PhD student at Univ. of Pisa)	
○ M. Piarulli, University of Milano-Bicocca, Master's thesis, 100%	2022-2023
(now PhD student at Univ. of Toulouse)	
○ A. Spadaro, University of Milano-Bicocca, Master's thesis, 100%	2021-2022
(now PhD student at Univ. of Milano-Bicocca)	
○ A. Carzaniga, University of Milano-Bicocca, Master's thesis, 100%	2021-2022
○ A. Geminardi, University of Milano-Bicocca, Master's thesis, 100%	2021-2022
(now PhD student at Univ. of Pavia)	
○ E. Finch, University of Birmingham, Year 4 project, 50%	2018
○ V. Spasova, University of Birmingham, Year 4 project, 50%	2018

BSc student supervision:

○ H. P. G. Carabajo, University of Milano-Bicocca, Bachelor's thesis, 100%	2023-2024
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Taught classes

	Acad.Year
○ Inference models in LISA , PhD course, Univ. of Trento (Italy).	2026
○ Current and future challenges in GW astronomy , PhD course, Milano-Bicocca (Italy).	2023
Content: Introduction to gravitational-wave data analysis. Space-based and ground-based interferometers. Source populations: compact object binaries, stochastic backgrounds.	
○ Mathematical physics and gravity (MAF900), PhD course, Univ. of Stavanger (Norway).	2023
Content: Introduction to gravitational-wave data analysis. Space-based interferometers. Source populations: compact object binaries, stochastic backgrounds. Signal modeling: galactic and extragalactic compact object binaries astrophysical and cosmological stochastic backgrounds. Signal detection and parameter estimation: frequentist and Bayesian approaches. Advanced stochastic sampling techniques.	
○ Lecture in Astrostatistics (F5802Q014/20), Astrophysics MSc, Univ. of Milano-Bicocca (Italy)	2022
Content: Source populations: compact object binaries, stochastic backgrounds. Introduction to gravitational-wave data analysis. Space-based interferometers. Signal detection and parameter estimation: frequentist and Bayesian approaches.	

Tutoring

	Acad.Year
○ Python Computing Lab , Bachelor's degree in Physics, Univ. of Birmingham, UK	2017-2021
Content: Python programming, simulation of physical systems in celestial and classical mechanics, thermodynamics, electromagnetism. Data analysis.	
○ Maths for physicists , Bachelor's degree in Physics, Univ. of Birmingham, UK	2017-2019
Content: Linear algebra, differential and integral calculus, differenzial equations Groups and representations theory	
○ Physics and communication skills , Master's degree in Physics, Univ. of Birmingham, UK.	2019
Content: $\text{\LaTeX}$ basics. Drafting of lab reports, publications, seminars	

Academic service, editorial and research responsibilities

International collaboration responsibilities

○ LISA Consortium AstroWG coordinator, project "Updatable predictions for stBBH in LISA"	2025-2026
○ PAQA co-chair of the Coordination Unit L2D (Global Fit, ESA LISA Project Office)	2024-2026
○ DDPC-Italy Lead of the LISA DDPC national coordination initiative	2024-2026

Journal referee

- Physical Review Letters
- The Astrophysical Journal Letters
- The Open Journal of Astrophysics
- NASA Technology Transfer Program
- IoP Classical and Quantum Gravity
- Institute of Physics Trusted Reviewer Excellence program
- Physical Review D
- Physics Letters B
- Journal of Cosmology and Astroparticle Physics
- Monthly Notices of the Royal Astronomical Society
- Institute of Physics Machine Learning: Science and Technology

Editorial responsibilities

- Editorial board Lensing working group in the LIGO, Virgo, KAGRA collaboration 2023
- Co-editor of a Living Review in relativity issue on “LISA data analysis” 2022-2023

Conference organizer

- *LISA Without Frontiers Workshop Sprint Session Coach*, Sexten, IT. 2026
- *LISA Distributed Data Processing Center June Workshop*, Milan, IT. 2025
- *IFPU focus week on “Emerging methods in GW population inference”*, Trieste, Italy. 2024
- *LISA Astrophysics Working Group Conference*, Birmingham, UK. 2022
- *Gravitational-wave populations: what's next?*, Milan, Italy. 2023
- *Gravitational-wave Excellence Alliance Training (GrEAT) PhD school*, Birmingham, UK. 2019
- *Gravitational-wave Open Science Center First Open Data Workshop*, (online) 2019

Professional recognition and service

- National Scientific Habilitation (ASN) as associated professor. 2023
in Astrophysics, Cosmology and Space Science (GSD 02/PHYS-05 - PHYS-05/A)
- French Qualification for teaching in Higher Education in Astrophysics (Sec.34), 2023
French Ministry of higher education and research (qualification no.23234388826).
- LSC Academic Advisory Committee. 2019-2021

Memberships

- LISA Distributed Data Processing Center, full member. since 2024
- LISA Consortium, core member. since 2018
- Italian Center for Supercomputing (ICSC). since 2021
- TEONGRAV National Initiative (Gravity Theory) since 2021
Italian National Institute for Nuclear Physics (INFN).
- LIGO, Virgo, Kagra Collaboration, full member. since 2017
- Italian Society of General Relativity and Gravitational Physics (SIGRAV) since 2021
- Istituto Nazionale di Astrofisica (INAF) since 2021
- American Physical Society (APS), member.
- Italian Physical Society (SIF), member 2021
- Royal Astronomical Society (RAS), fellow. 2018-2021

Date: June 8, 2026